Problem Set 3

1. Warm up: In S_4, show that $s_1s_2s_1 \leq s_2s_1s_3s_2$ and that the elements $s_1s_2s_3$ and $s_2s_1s_3s_2$ are incomparable in Bruhat order.

2. Prove that there is a reflection $t \in T$ such that $u = tv$ if and only if there is a reflection $t' \in T$ such that $u = vt'$.

3. Prove, for $w \in W$, that the total number of incoming edges to and outgoing edges from w in the Bruhat graph is $|T|$.

4. How many total edges are there in the Bruhat graph of S_n? How many total cover relations are in the weak order on S_n?

5. Fix $1 \leq k < n$. A permutation is called k**-Grassmannian** if it is equal to e or has s_k as its unique descent.
   (a) Show that taking the set $X_\pi = \{\pi(1), \ldots, \pi(k)\}$ gives a bijection between k-Grassmannian permutations and subsets of $[n]$ of size k.
   (b) Show that k-Grassmannian permutations π_1, π_2 satisfy $\pi_1 \leq \pi_2$ if and only if $X_{\pi_1} \leq X_{\pi_2}$ in Gale order.
   (c) Find a bijection between subsets of $[n]$ of size k and partitions that fit in a box of size $k \times (n-k)$. Show that $X \leq X'$ in Gale order if and only if the corresponding partitions λ, λ' satisfy $\lambda_i \leq \lambda'_i$ for all $1 \leq i \leq k$.
   (d) Bonus: show that there is an analog of the Bruhat decomposition which says

$$\mathrm{GL}_n(\mathbb{C}) = \bigsqcup_{w \text{ is } k\text{-Grassmannian}} BwP_{n,k},$$

   where $P_{n,k}$ is the group of matrices of the form $\begin{pmatrix} A & D \\ \mathbf{0} & C \end{pmatrix}$, where $A \in \mathrm{GL}_k(\mathbb{C})$ and $C \in \mathrm{GL}_{n-k}(\mathbb{C})$.

   You saw in the Schubert calculus problem set that the Grassmannian $\mathsf{Gr}(k,n)$ is isomorphic to $\mathrm{GL}_n(\mathbb{C})/P_{n,k}$. Define the Schubert cell Ω'_λ to be $BwP_{n,k}/P_{n,k}$, where w is the k-Grassmannian permutation associated to λ. (Note this is not quite the same definition as in the Schubert calculus lectures; the Ω_λ defined there is isomorphic to our Ω'_{λ^c}.) It follows from what you have shown in this problem (and the compactness of the flag variety) that $\Omega'_{\lambda_1} \subseteq \overline{\Omega'_{\lambda_2}}$ if and only if $\lambda_1 \leq \lambda_2$.

6. Show that the length generating function of S_n

$$f(q) = \sum_{w \in S_n} q^{\ell(w)}$$

   factors as

$$f(q) = (1+q)(1+q+q^2)(1+q+q^2+q^3)\cdots(1+q+\cdots+q^{n-1}).$$

   The polynomial $1 + q + \cdots + q^{k-1}$ is called a q**-integer** or **quantum integer**, denoted $[k]_q$. (This is because when you set $q = 1$ you get the classical integer k.) The function $f(q)$ is called the q**-factorial**, denoted $[n]_q!$.

7. Prove that if $u \le v$ and $\ell(v) = \ell(u) + 2$, then there are exactly two elements x such that $u < x < v$. Another way of phrasing this is to say that the **Bruhat interval** $[u, v]$ has the Hasse diagram

$$\begin{tikzcd} & v \arrow[dl, dash] \arrow[dr, dash] & \\ x \arrow[dr, dash] & & y \arrow[dl, dash] \\ & u & \end{tikzcd} .$$

8. Prove that, in the symmetric group, $u \le v$ if and only if $w_0 u w_0 \le w_0 v w_0$ (this is in fact true for all finite Coxeter groups).

9. In the symmetric group, prove the **Tableau criterion**. This says that, to check that $u \le v$, we just need to check that $\{u(1), \dots, u(k)\} \le \{v(1), \dots, v(k)\}$ in Gale order for every k such that s_k is a descent of u.

10. In the symmetric group S_n, let's list all the transpositions as $t_1, t_2, \dots, t_{\binom{n}{2}}$ in lexicographic order, so that $t_1 = (1,2), t_2 = (1,3), \dots, t_{n-1} = (1,n), t_n = (2,3), \dots, t_{\binom{n}{2}} = (n-1, n)$. Let $u < v$ be two permutations related by Bruhat order. Define a sequence of permutations by the following algorithm: set $u_0 = u$. At step i, do the following:
    - If $u_{i-1} < t_i u_{i-1} \le v$, then set $u_i = t_i u_{i-1}$.
    - Otherwise, set $u_i = u_{i-1}$.

    Prove that $u_{\binom{n}{2}} = v$. Moreover, show for all i that either $u_i = u_{i-1}$ or $\ell(u_i) = \ell(u_{i-1}) + 1$.

11. Assume $u \to v$ is an edge of the Bruhat graph such that $\ell(v) = \ell(u) + 3$. Prove that the Bruhat interval $[u, v] := \{x \in W \mid u \le x \le v\}$ has 6 elements, with Hasse diagram

$$\begin{tikzcd} & v \arrow[dl, dash] \arrow[dr, dash] & \\ x_2 \arrow[d, dash] \arrow[drr, dash] & & y_2 \arrow[d, dash] \arrow[dll, dash] \\ x_1 \arrow[dr, dash] & & y_1 \arrow[dl, dash] \\ & u & \end{tikzcd} .$$

    Bonus: Prove the converse.

12. For each $w \in S_n$, show that there is an $\ell(w)$-dimensional subgroup U_w of B such that the map $\phi : U_w \to \mathrm{GL}_n(\mathbb{C})/B$ sending $u \in U_w$ to the coset uwB is bijective. (A consequence is that the Schubert cell BwB/B in the flag variety has complex dimension $\ell(w)$.)

13. The **Mobius function** of a poset P assigns to each pair $x \le y$ an integer defined by

$$\mu(x, y) := \begin{cases} 1 & \text{if } x = y \\ -\sum_{x \le z < y} \mu(x, z) & \text{if } x < y \end{cases} .$$

    Show that Bruhat order on W is an **Eulerian poset**, meaning $\mu(u, v) = (-1)^{\ell(v) - \ell(u)}$ for all $u \le v$.